

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI



IDC216 (Statistics for Scientists) / Monsoon 2025 / End Semester Exam

Maximum Marks : 30

November 25, 2025

- All questions will be considered in the evaluation.
- Use of calculators is allowed. However, exchange of calculators and use of mobile apps is not permitted.
- Organize your work, in a reasonably neat and coherent way. Work scattered all over the pages without a clear ordering may be missed by the evaluator. Illegible work will not receive credit.
- Mysterious or unsupported answers will not receive credit. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations *might* still receive partial credit.

1. A researcher collects monthly average air quality data for Delhi for one calendar year. The data for Particulate Matter ($PM_{2.5}$, in $\mu g/m^3$), Sulfur Dioxide (SO_2 , in parts per billion (ppb)), and the overall Air Quality Index (AQI) are presented in the table below.

Month	$PM_{2.5}$	SO_2	AQI
1 Jan	184	36	310
2 Feb	166	35	305
3 Mar	135	30	235
4 Apr	105	25	188
5 May	95	22	190
6 Jun	75	29	160
7 Jul	60	19	130
8 Aug	65	21	135
9 Sep	85	24	175
10 Oct	125	26	240
11 Nov	155	49	306
12 Dec	193	33	358

- Determine, with a supporting calculation, which pollutant is a stronger predictor of the AQI in this dataset, $PM_{2.5}$ or SO_2 ? Justify the reasoning behind the choice of calculation. [3+1]
2. The heights of students at an institute are known to be normally distributed with a mean of 170 cm and a standard deviation of 8 cm. A researcher plans to select a sample of 16 students, and wants the average height of this sample to fall in the top 10% of all possible sample mean heights for samples of this size.

(a) What is the minimum sample mean height required for the sample to be in the top 10% of all sample means? [3]

(b) State the theorem used in part (a), and explain why it is applicable in this situation. [1]

3. A study investigates the relationship between the concentration of a catalyst (in mg/L, x) and the reaction rate (in mol/s, y) in a chemical process. The following data were collected:

x	10	15	20	25	30	35
y	1.2	1.8	2.5	3.1	3.6	2.9

(a) Fit a least squares regression line of y on x . Predict the reaction rate when the catalyst concentration is 22 mg/L. [3]

(b) The data point at $x = 35$ is suspected to be an outlier. If this point is removed, would you expect the slope of the regression line to increase or decrease? Provide a brief conceptual justification for your answer. [1]

4. A plant scientist claims that a new plant variety has a mean seed germination time of less than 5 days. To test this, she plants 25 seeds and finds a sample mean germination time of 4.7 days with a sample standard deviation of 0.8 days.

(a) Formulate the null and alternative hypotheses. [1]

(b) Perform a t -test at a significance level of $\alpha = 0.05$. State your conclusion in the context of the problem. [3]

5. A manufacturer of the board game *Snakes and Ladders* advertises that all dice provided by them are perfectly fair. During the annual *IISER Mohali Snakes and Ladders Tournament*, where dice provided by this manufacturer are used, a vigilant participant becomes suspicious and records the outcomes of 240 rolls of one such die. The observations are shown below:

Face	1	2	3	4	5	6	Total
Frequency	48	35	45	32	50	30	240

Concerned that the die may be biased, this vigilant participant files a formal complaint with the *Snakes and Ladders Authority of India*. You are appointed as the arbitrator to investigate this matter.

(a) Conduct a statistical investigation to assess whether the data provide evidence that the die is fair. [3]

(b) Regardless of the outcome of part (a), propose a reasonable and justifiable follow-up plan to ensure a fair resolution of the dispute. [1]

6. Read the provided scientific correspondence from Rauscher et al. (Nature, 1993), *Music and spatial task performance*. Examine this study from a statistical perspective by addressing the following:

(a) Identify the independent variable(s) and dependent variable(s) in this experimental design. [2]

(b) The study included 36 college students. Discuss concerns related to this. [2]

(c) The authors report $F_{2,35} = 7.08$, $p = 0.002$ for the repeated-measures ANOVA. What null hypothesis is being tested, and how should this result be interpreted? [2]

(d) Identify two potential confounding variables that might not be adequately controlled for in this study. [2]

(e) A news headline based on this study reads: "*Mozart's Music Makes You Smarter!*" As a statistically informed citizen, critique this statement in light of this study. [2]

Table 1: t-Distribution Critical Values

df	Significance Level α (one-tailed)				
	0.10	0.05	0.025	0.01	0.005
16	1.337	1.746	2.120	2.583	2.921
20	1.325	1.725	2.086	2.528	2.845
24	1.318	1.711	2.064	2.492	2.797
28	1.313	1.701	2.048	2.467	2.763

Table 2: Chi-Square (χ^2) Distribution Critical Values

df	Significance Level α				
	0.99	0.95	0.05	0.01	0.001
1	0.000	0.004	3.841	6.635	10.828
2	0.020	0.103	5.991	9.210	13.816
3	0.115	0.352	7.815	11.345	16.266
4	0.297	0.711	9.488	13.277	18.467
5	0.554	1.145	11.070	15.086	20.515
6	0.872	1.635	12.592	16.812	22.458

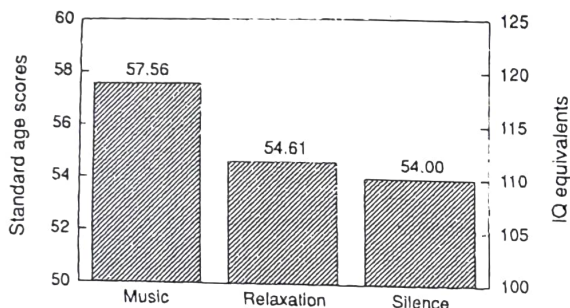
Table 3: Standard Normal Distribution Table for $\Phi(z) = P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

Music and spatial task performance

SIR—There are correlational¹, historical² and anecdotal³ relationships between music cognition and other 'higher brain functions', but no causal relationship has been demonstrated between music cognition and cognitions pertaining to abstract operations such as mathematical or spatial reasoning. We performed an

spatial IQ scores of 119, 111 and 110, respectively. Thus, the IQs of subjects participating in the music condition were 8–9 points above their IQ scores in the other two conditions. A one-factor (listening condition) repeated measures analysis of variance (ANOVA) performed on SAS revealed that subjects



Standard age scores for each of the three listening conditions.

Testing procedure. In the music condition, the subject listened to 10 min of the Mozart piece. The relaxation condition required the subject to listen to 10 min of relaxation instructions designed to lower blood pressure. The silence condition required the subject to sit in silence for 10 min. One of three abstract reasoning tests taken from the Stanford-Binet intelligence scale⁴ was given after each of the listening conditions. The abstract/spatial reasoning tasks consisted of a pattern analysis test, a multiple-choice matrices test and a multiple-choice paper-folding and cutting test. For our sample, these three tasks correlated at the 0.01 level of significance. We were thus able to treat them as equal measures of abstract reasoning ability.

Scoring. Raw scores were calculated by subtracting the number of items failed from the highest item number administered. These were then converted to SAS using the Stanford-Binet's SAS conversion table of normalized standard scores with a mean set at 50 and a standard deviation of 8. IQ equivalents were calculated by first multiplying each SAS by 3 (the number of subtests required by the Stanford-Binet for calculating IQs). We then used their area score conversion table, designed to have a mean of 100 and a standard deviation of 16, to obtain SAS IQ equivalents.

experiment in which students were each given three sets of standard IQ spatial reasoning tasks; each task was preceded by 10 minutes of (1) listening to Mozart's sonata for two pianos in D major, K488; (2) listening to a relaxation tape; or (3) silence. Performance was improved for those tasks immediately following the first condition compared to the second two.

Thirty-six college students participated in all three listening conditions. Immediately following each listening condition, the student's spatial reasoning skills were tested using the Stanford-Binet intelligence scale⁴. The mean standard age scores (SAS) for the three listening conditions are shown in the figure. The music condition yielded a mean SAS of 57.56; the mean SAS for the relaxation condition was 54.61 and the mean score for the silent condition was 54.00. To assess the impact of these scores, we 'translated' them to

performed better on the abstract/spatial reasoning tests after listening to Mozart than after listening to either the relaxation tape or to nothing ($F_{2,35} = 7.08$; $P = 0.002$). The music condition differed significantly from both the relaxation and the silence conditions (Scheffe's $t = 3.41$, $P = 0.002$; $t = 3.67$, $P = 0.0008$, two-tailed, respectively). The relaxation and silence conditions did not differ ($t = 0.795$; $P = 0.432$, two-tailed). Pulse rates were taken before and after each listening condition. A two-factor (listening condition and time of pulse measure) repeated measures ANOVA revealed no interaction or main effects for pulse, thereby excluding arousal as an obvious cause. We found no order effects for either condition presentation or task, nor any experimenter effect.

The enhancing effect of the music condition is temporal, and does not extend beyond the 10–15-minute period during which subjects were engaged in each spatial task. Inclusion of a delay period (as a variable) between the music listening condition and the testing period would allow us quantitatively to determine the presence of a decay constant. It would also be interesting to vary the listening time to optimize the enhancing effect, and to examine whether other measures of general intelligence (verbal reasoning, quantitative reasoning and short-term memory) would be similarly facilitated. Because we used only one musical sample of one composer, various other compositions and musical styles

should also be examined. We predict that music lacking complexity or which is repetitive may interfere with, rather than enhance, abstract reasoning. Also, as musicians may process music in a different way from non-musicians, it would be interesting to compare these two groups.

Frances H. Rauscher

Gordon L. Shaw*

Katherine N. Ky

Center for the Neurobiology of Learning and Memory,

University of California, Irvine, California 92717, USA

* Also at Department of Physics.

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3. Cranberg, L. D. & Albert, M. L. in *The Exceptional Brain* (eds Opler, L. K. & Fein, D.) 156 (Guilford, New York, 1988).
4. Thorndike, R. L., Hagen, E. P. & Sattler, J. M. *The Stanford-Binet Scale of Intelligence* (Riverside, Chicago, 1986).

MyoD and c-fos expression

SIR—Trousche *et al.*¹ have reported that the down regulation of *c-fos* expression during muscle cell differentiation may result from the binding of the helix-loop-helix (HLH) proteins to a CANNTG motif, or E-box, that occurs within the *c-fos* serum response element (SRE), thereby excluding the binding of the serum response factor (SRF) to this element. We investigated the interaction of HLH proteins with the *c-fos* SRE when molecular clones for E12 were first isolated by screening a phage expression library with the SRE probe². We estimated that the dissociation constant for the myogenin/E12-SRE complex was 10^{-8} – 10^{-9} M⁻¹ by comparing the relative effectiveness of E-box elements from different genes to compete for binding in the electrophoretic mobility-shift assays. The relatively low affinity of myogenin/E12 for the SRE could result from differences in nucleotide identities at the internal dinucleotide and flanking sequences between the *c-fos* E-box and the consensus HLH binding site^{3,4}. By contrast, the dissociation constants for the SRE-SRF or the SRE-SRF/p62^{TCF} complexes have been reported to be 5×10^{-10} and $\leq 10^{-11}$ M⁻¹, respectively⁵. Collectively, these data indicate that it is unlikely that HLH proteins alone can significantly compete with SRF for binding to DNA *in vivo*.

It has been reported in other studies that the *c-fos* SRE is either equally active in muscle and non-muscle cells⁶, or that it activates muscle-specific expression⁷ when situated upstream from a minimal promoter. Further, a comparison of the

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